

Metacognitive Illusions in Reasoning: Misleading Surface Cues Influence Monitoring and Control

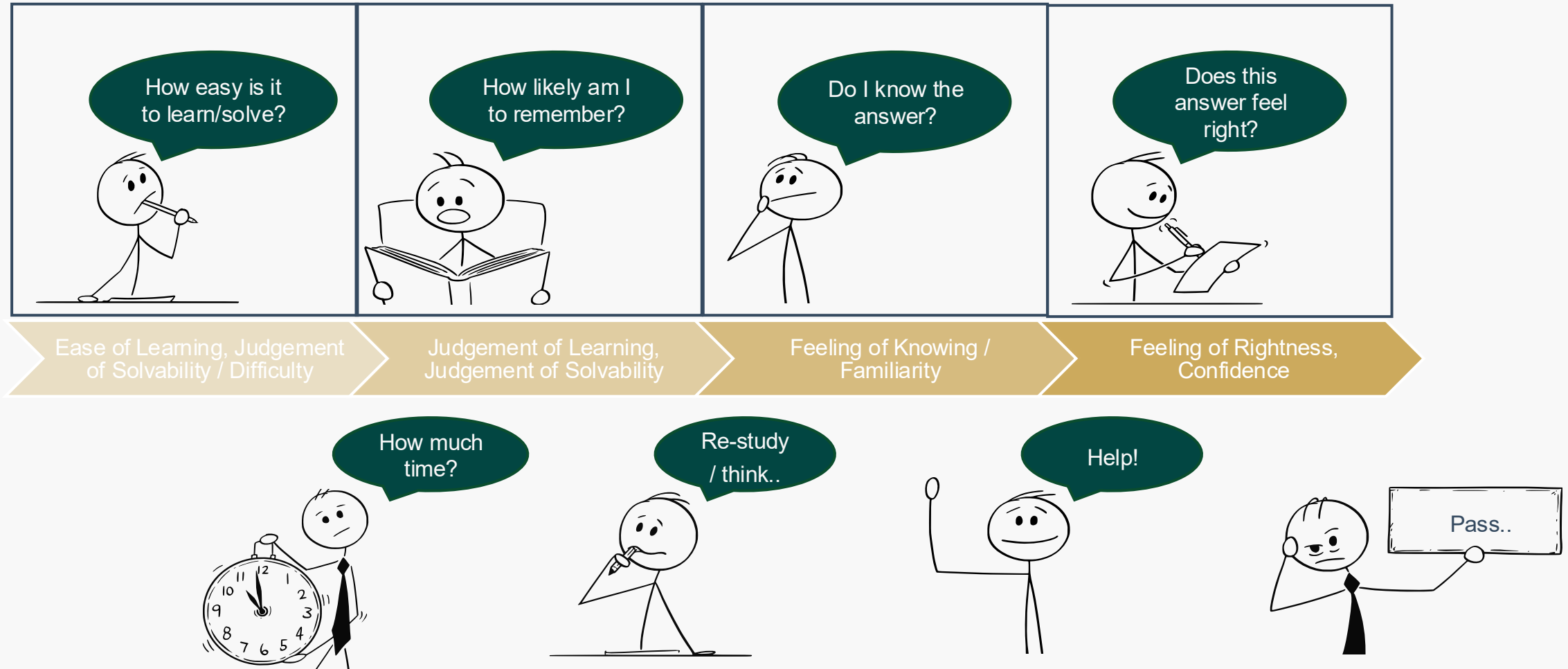


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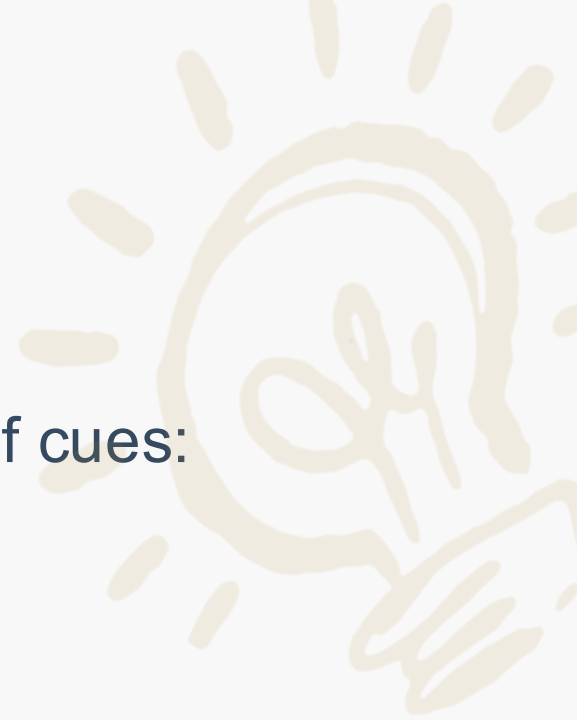
Metacognition



Monitoring is cue-based

Cue-utilization framework makes a distinction between 3 types of cues:

- Intrinsic cues (e.g. font size, believability, pronounceability)
- Extrinsic cues (e.g. test format, training)
- Mnemonic cues (e.g. ease of processing, ease of retrieval)



Inferences underlying monitoring judgements

Experience-based / non-analytic / implicit inferences

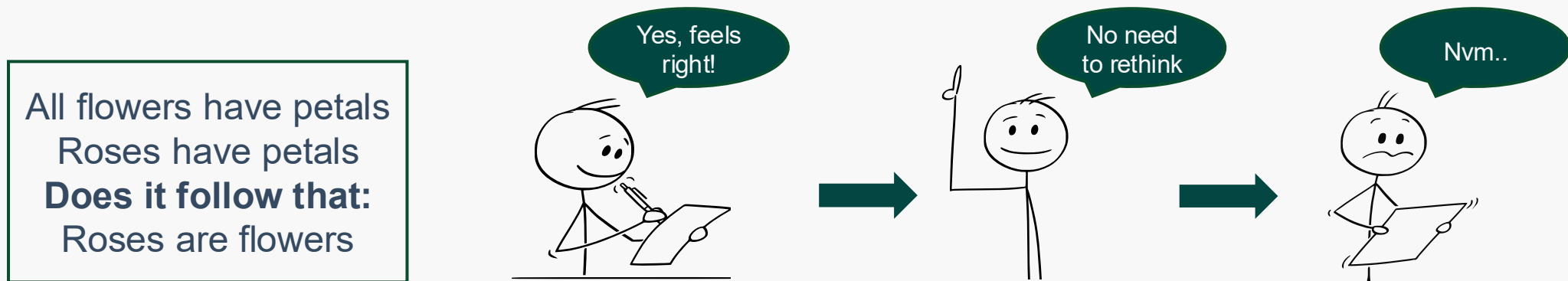
Theory-based / analytic / explicit inferences

All flowers have petals
Roses have petals
Does it follow that:
Roses are flowers



Metacognitive illusions & control

- **Metacognitive accuracy** is critical because monitoring is expected to inform control
- **Metacognitive illusions** arise when there is a dissociation between monitoring and performance
- Inaccurate monitoring can **mislead control decisions**, resulting in **suboptimal performance**



Current research

- Studies have shown that various cues affect intermediate and retrospective monitoring judgements
- Very few studies have examined how people make prospective judgements and the link between these judgements and control decisions
- No studies have examined theory-based inferences in reasoning

Our research examines whether: (1) people make theory-based inferences from cues, and (2) the same inferences that affect monitoring also affect control decisions

Study 1: Do people have theories about logical reasoning?

- Participants make comparative difficulty judgement and then explain the basis of their judgements
- Manipulated surface cues: familiarity, conclusion believability
- Controlled for logical validity & structure* across pairs

Which logic problem is more likely to be correctly solved by hypothetical participants?

Problem A	Problem B
Some <u>composers</u> are <u>doctors</u> All <u>doctors</u> are <u>farmers</u> Does it follow that: Some <u>composers</u> are <u>farmers</u>	Some <u>trantoos</u> are <u>extrics</u> All <u>extrics</u> are <u>sufts</u> Does it follow that: Some <u>trantoos</u> are <u>sufts</u>

A is more likely to
be correctly
solved

A and B are
equally likely to be
correctly solved

B is more likely to
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Problem A	Problem B
All glissomae are whales All mammals are glissomae Does it follow that: <u>All mammals are whales</u>	All cyrptods are amphibians All frogs are cyrptods Does it follow that: <u>All frogs are amphibians</u>

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Problem A	Problem B
No bictoids are rabbits All insects are bictoids Does it follow that: <u>No rabbits are insects</u>	No bakers are pinders All bowdlers are bakers Does it follow that: <u>Some bowdlers are pinders</u>

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Problem A	Problem B
No cars are hydrovans Some hydrovans are Volvos Does it follow that: <u>Some Volvos are not cars</u>	No embossers are butchers All butchers are melderers Does it follow that: <u>Some embossers are not melderers</u>

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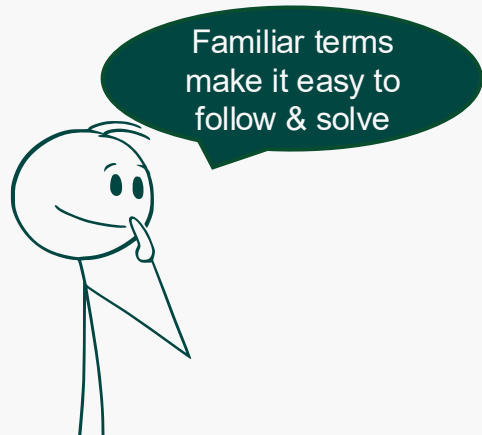
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Study 1: Do people have theories about familiarity?

Logic problems with familiar terms elicit higher confidence than nonword terms, despite not affecting performance

Hypothesis 1: Problems with familiar terms will be rated as easier to solve because they are processed more fluently



Which logic problem is more likely to be correctly solved by hypothetical participants?

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A and B are equally likely to be correctly solved

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Study 1: Do people have theories about familiarity?

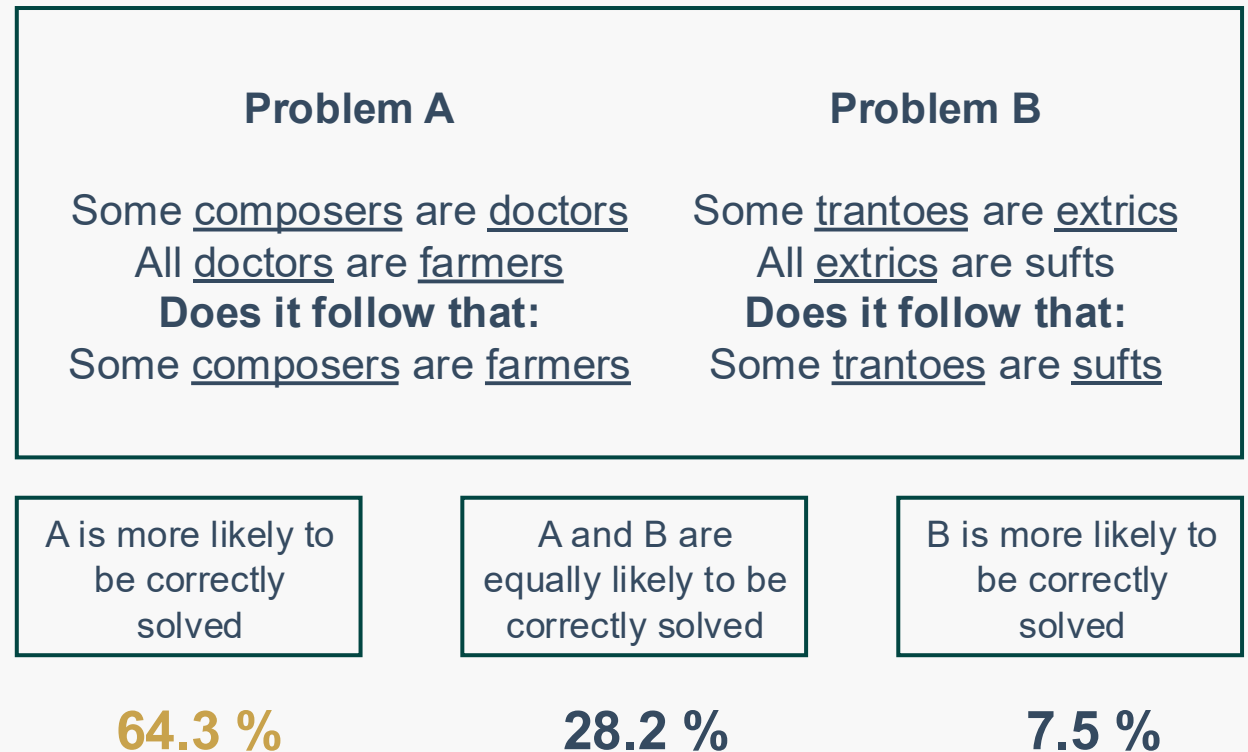
Logic problems with familiar terms elicit higher confidence than nonword terms, despite not affecting performance

Hypothesis 1: Problems with familiar terms will be rated as easier to solve because they are processed more fluently

“Unfamiliar words create an obstacle to the thought process”

“Familiar words making it easy to picture and understand”

Which logic problem is more likely to be correctly solved by hypothetical participants?

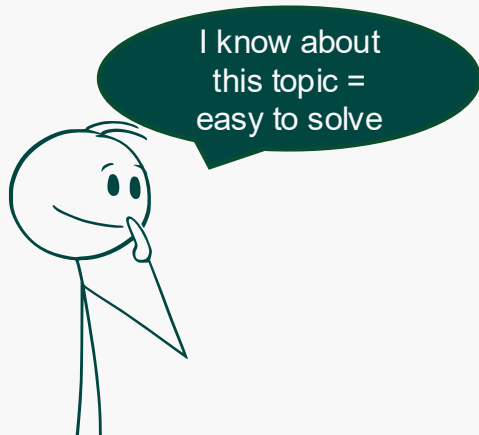


Study 1: Do people have theories about conclusion believability?

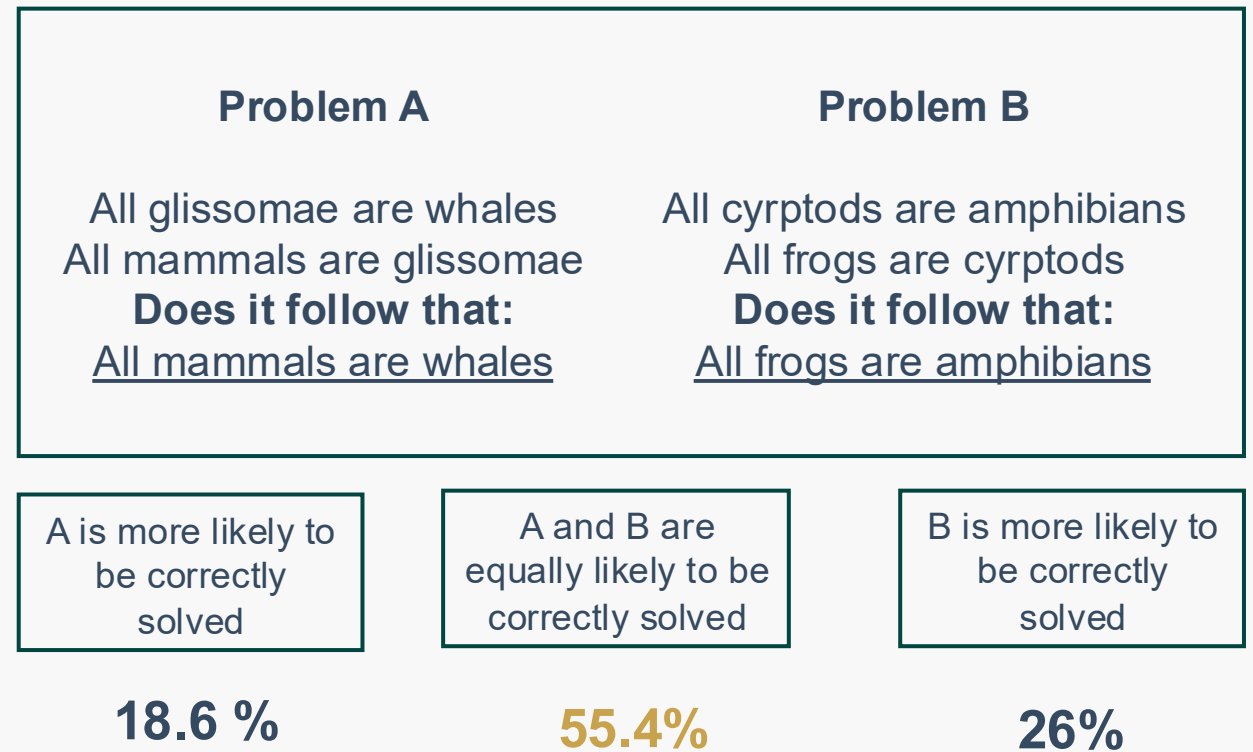
Conclusion believability interferes with performance on logic problems

H2a: *Believable conclusions would be rated easier than unbelievable ones*

H2b: *Both belief-based conclusions would be rated easier than neutral ones*



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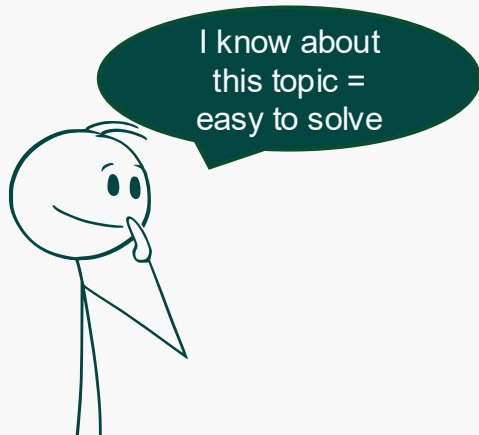


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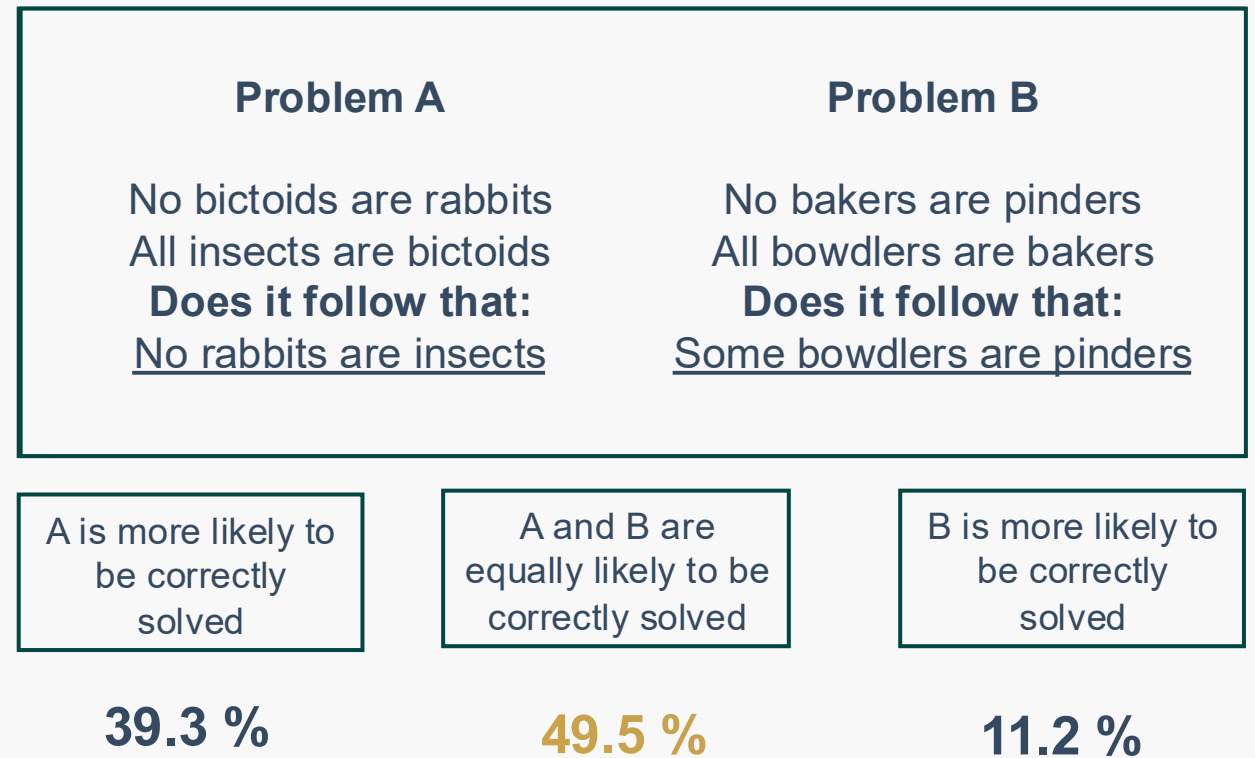
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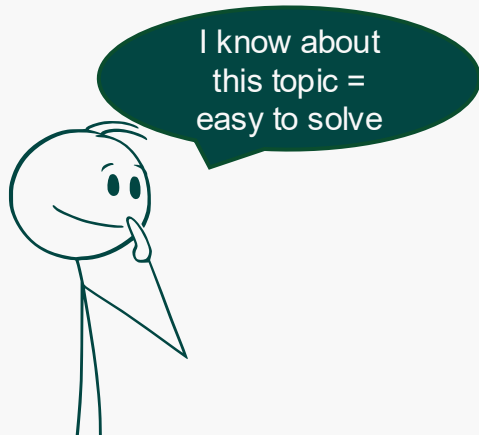


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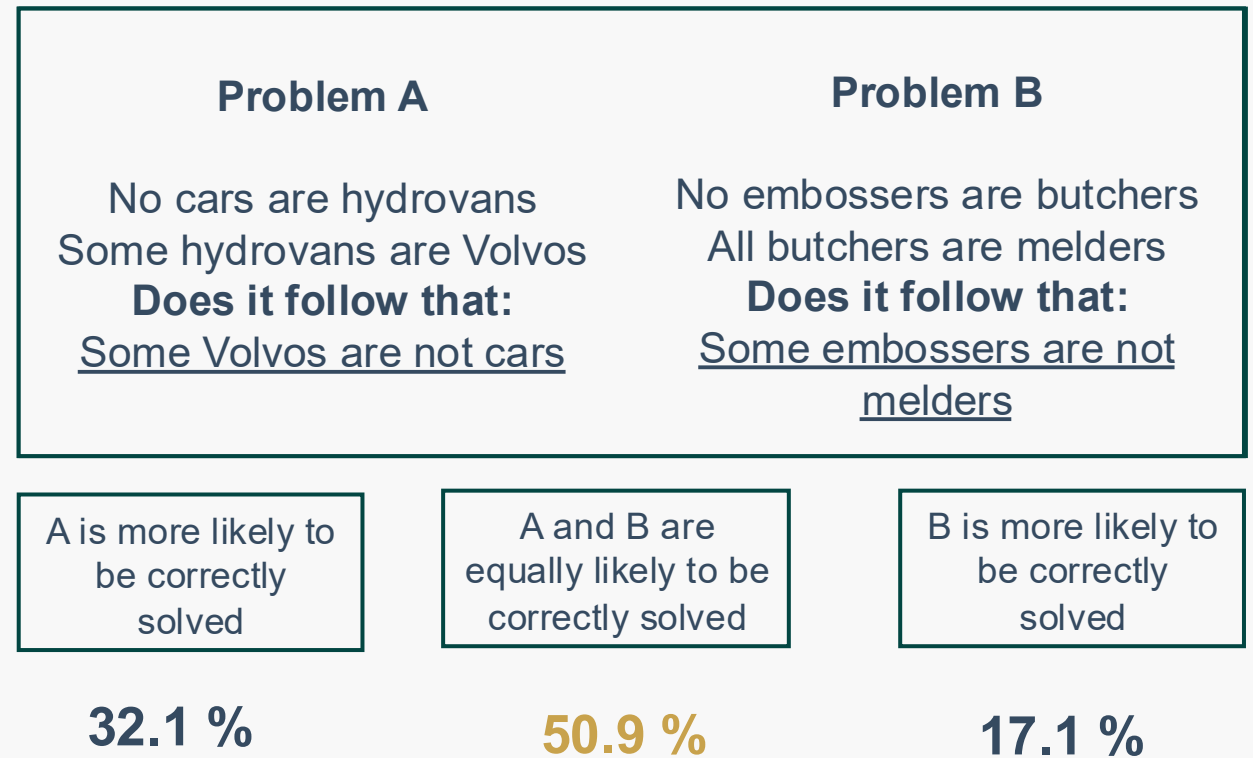
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Study 1 summary & Study 2

Study 1: Misleading theories about logical reasoning

Familiarity

- Misleading theory: Presence of familiar terms affect difficulty because they are easier to process
- Actual effect: Term familiarity *does not affect* logical reasoning

Conclusion believability

- No specific theory
- Actual effect: Conclusion believability *does affect* logical reasoning

Study 2: Do misleading cues & theories also mislead control decisions?

Are there systematic dissociations between JODs and actual performance?

Do these same cue manipulations directly affect control decisions?

- The same cues that distort monitoring can also mislead control decisions, resulting in ineffective problem-solving strategies

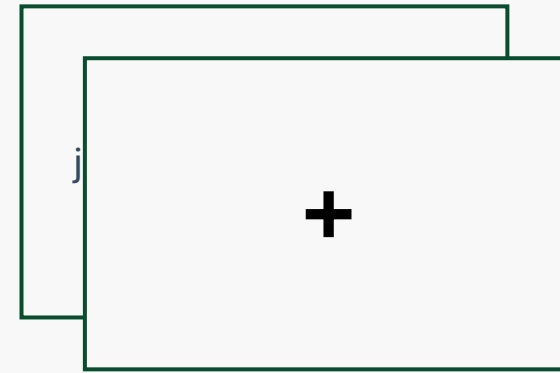
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1. Replicate Study 1 findings with single-item presentation to rule out comparative judgment artifacts
2. Compare prospective (JOD) vs. retrospective (confidence) judgments and their sensitivity to cues
3. Examine monitoring-accuracy dissociation by examining actual performance
4. Test causal effects of cues on control decisions (i.e. giving up)

You will be asked to judge the difficulty of the next problem.

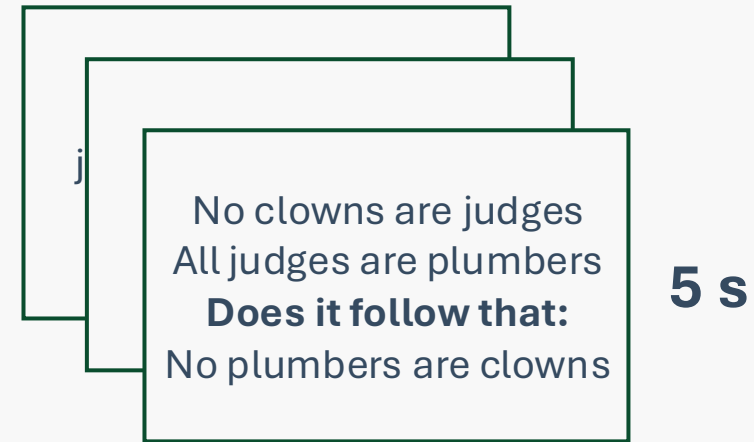
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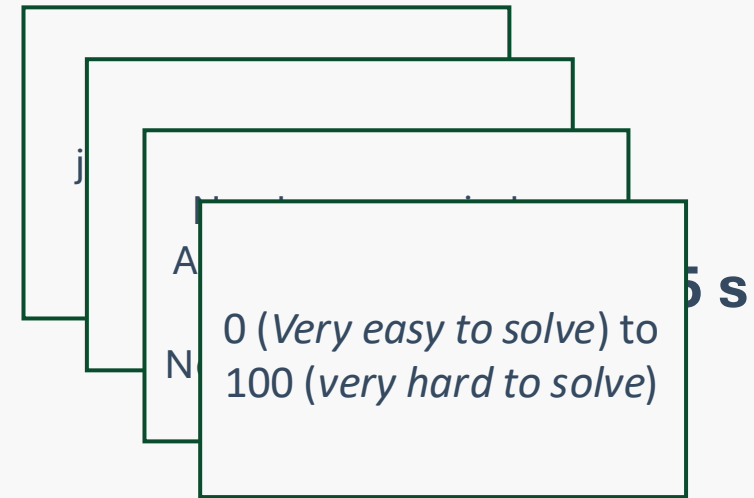
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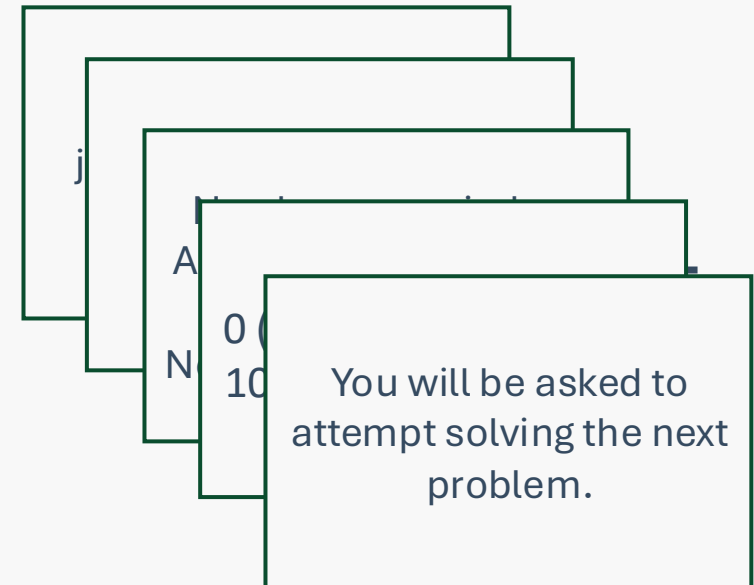
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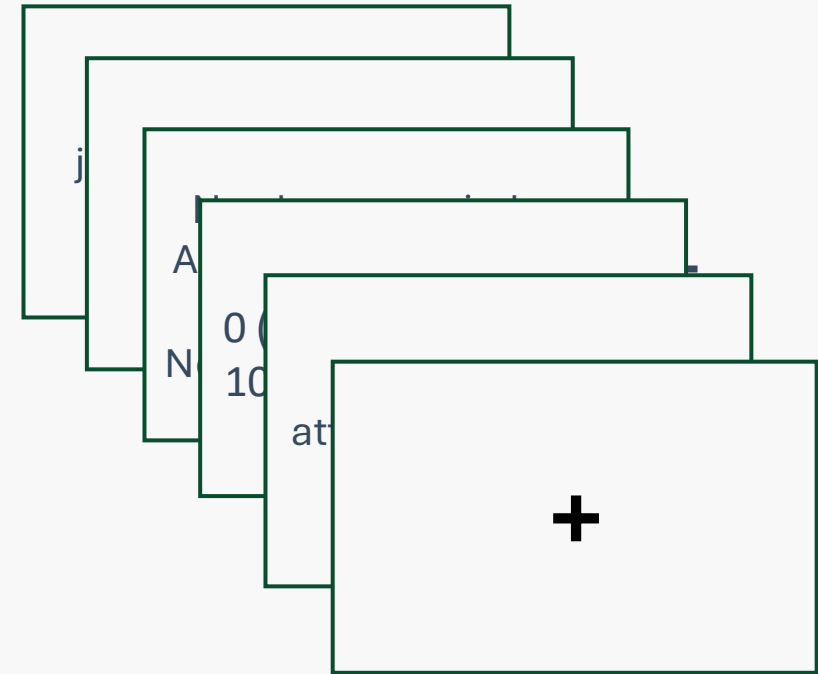
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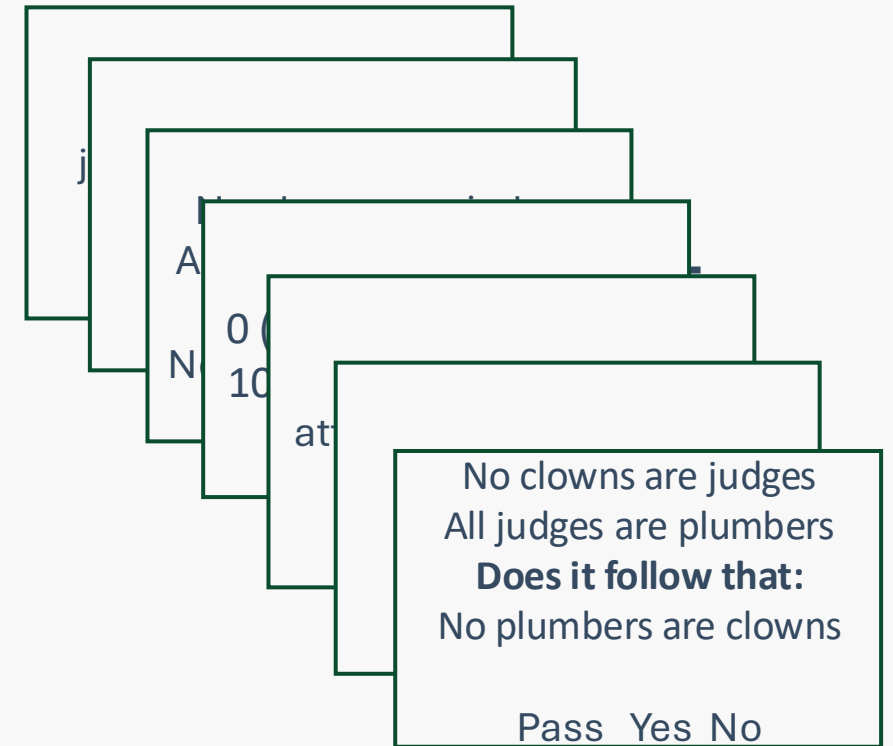
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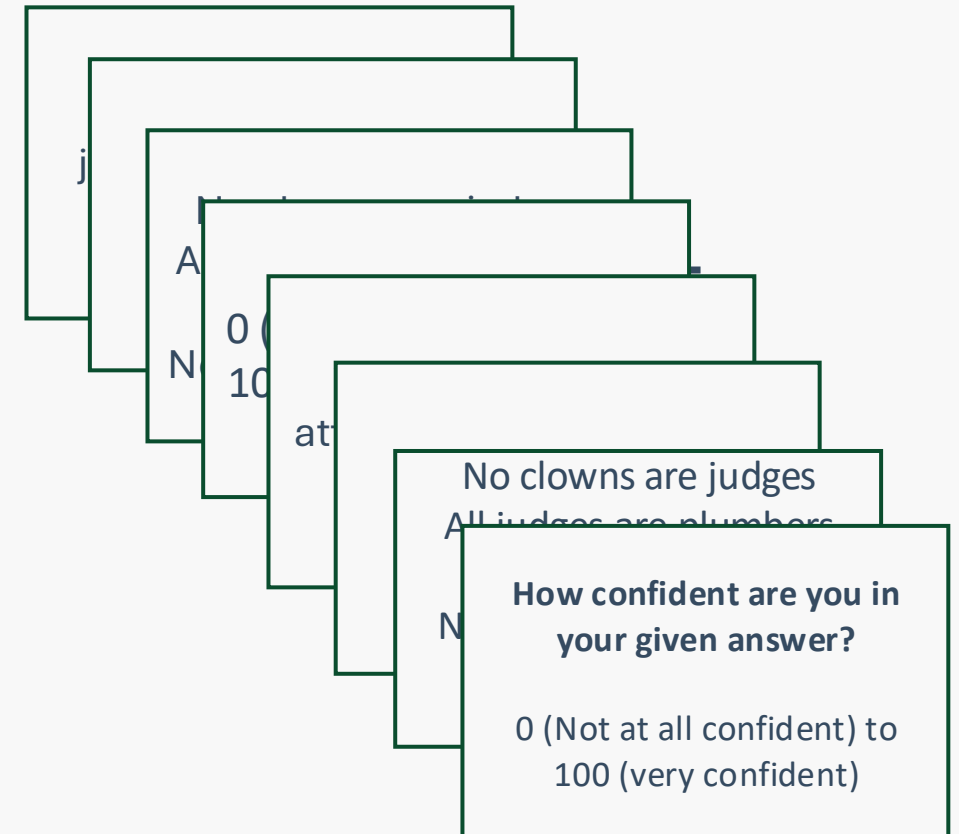
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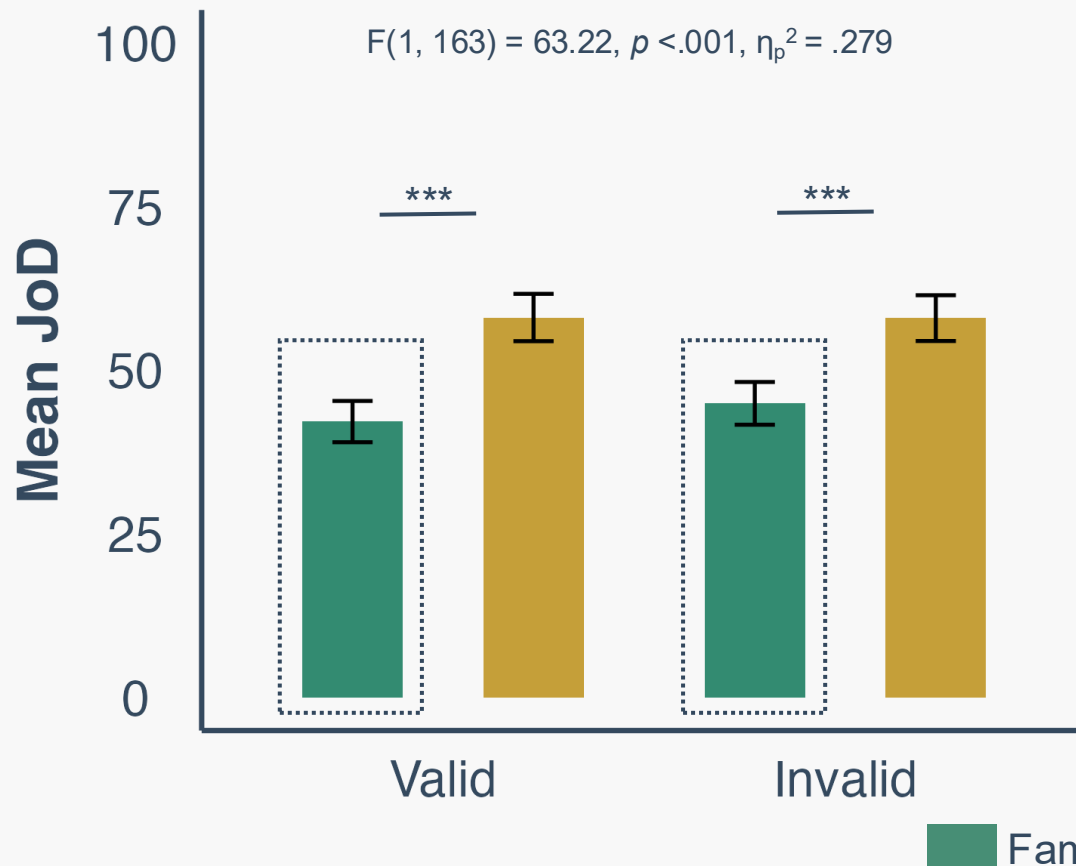
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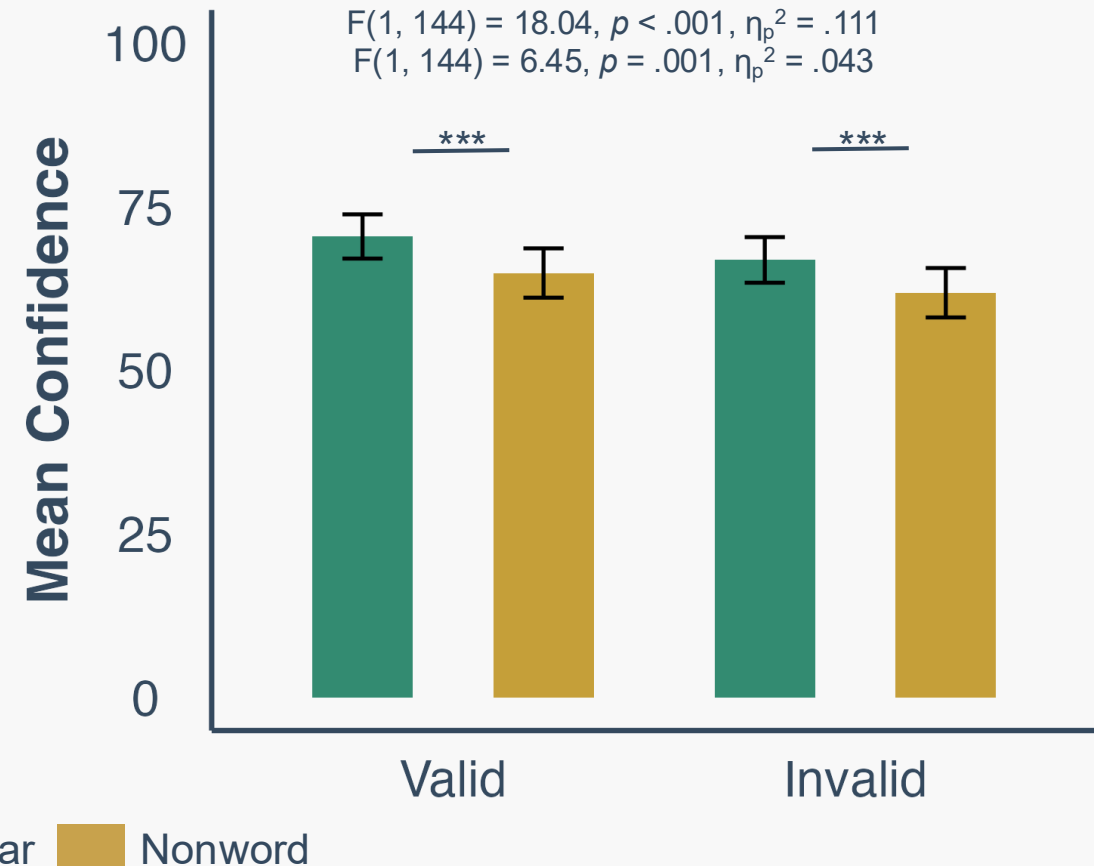


Study 2: Familiarity affects JOD & confidence

H1: Familiar terms → lower difficulty ratings

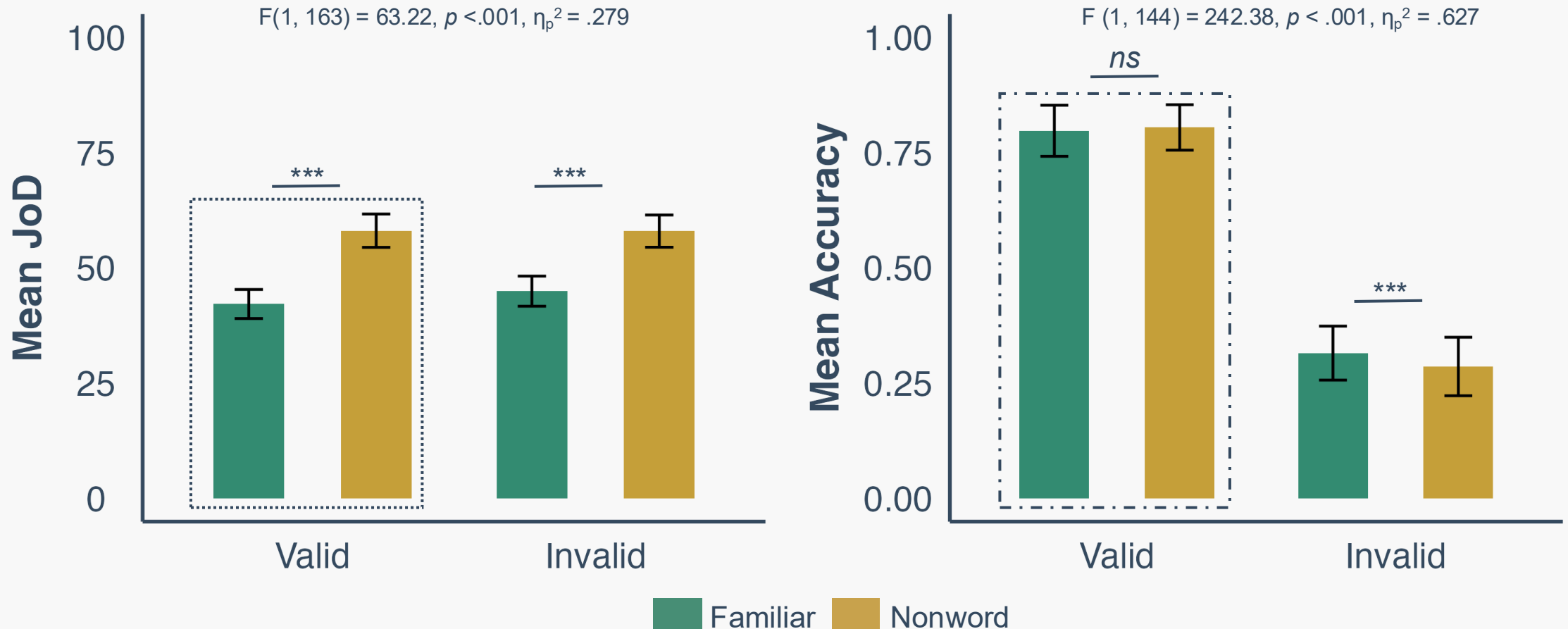


H2: Confidence tracks JODs (low difficulty → high confidence)



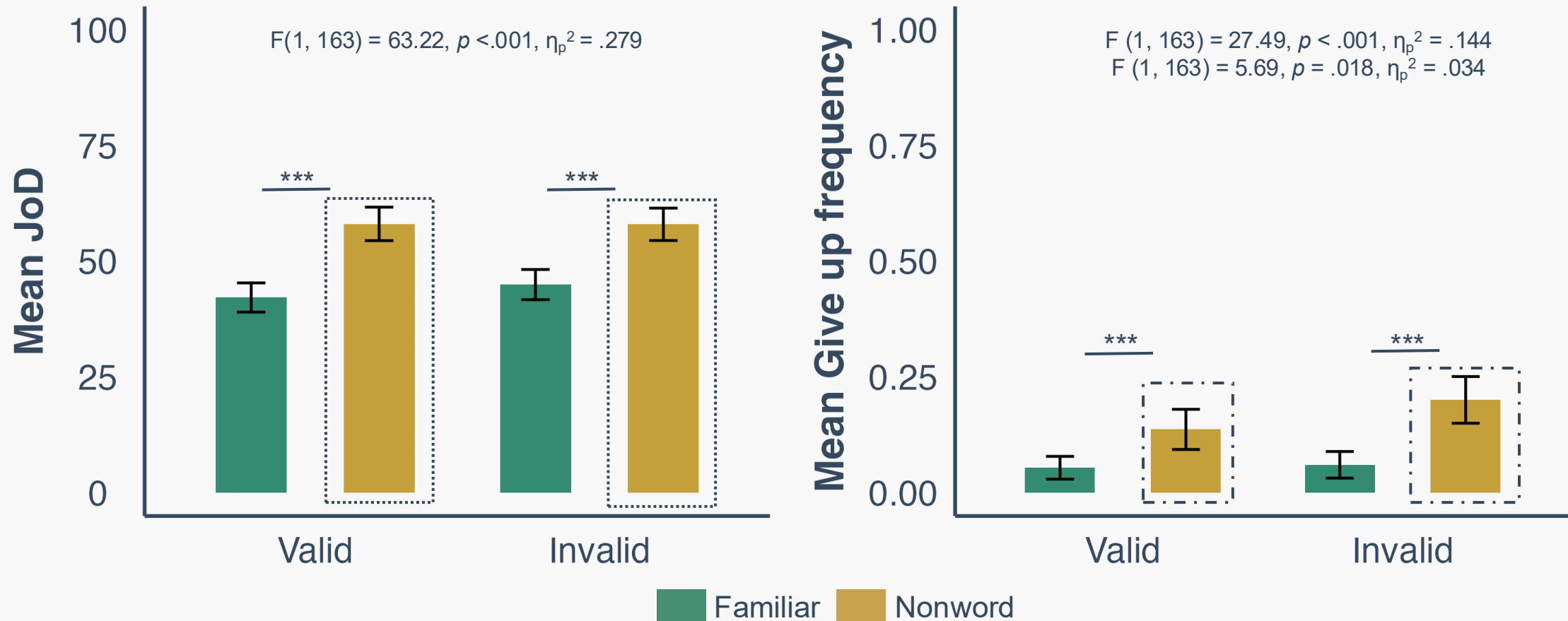
Study 2: Familiarity leads to metacognitive illusions

H3: JOD-Accuracy dissociation; familiarity affects JOD, while logical validity affect performance



Study 2: Familiarity affects control decisions

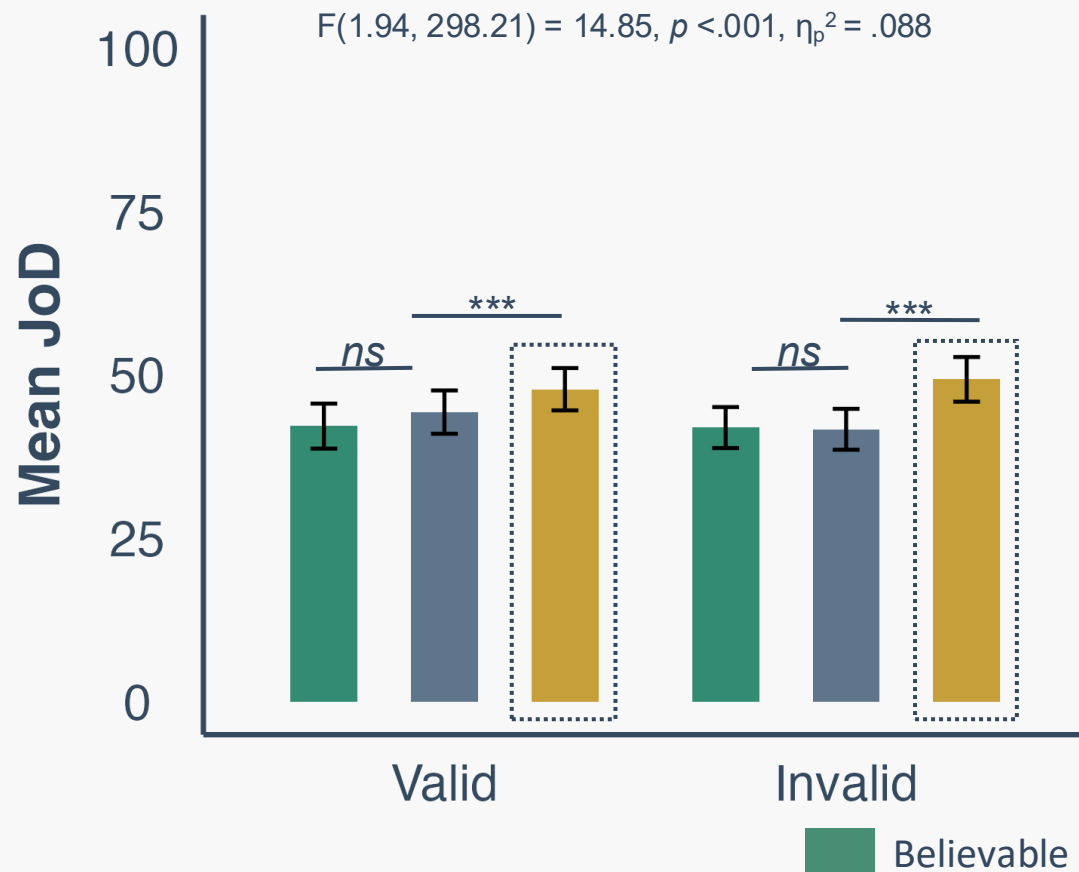
H4: The decision to give up will tracks JOD; people will give up more on problems with nonwords



Study 2: Believability affects JOD & confidence

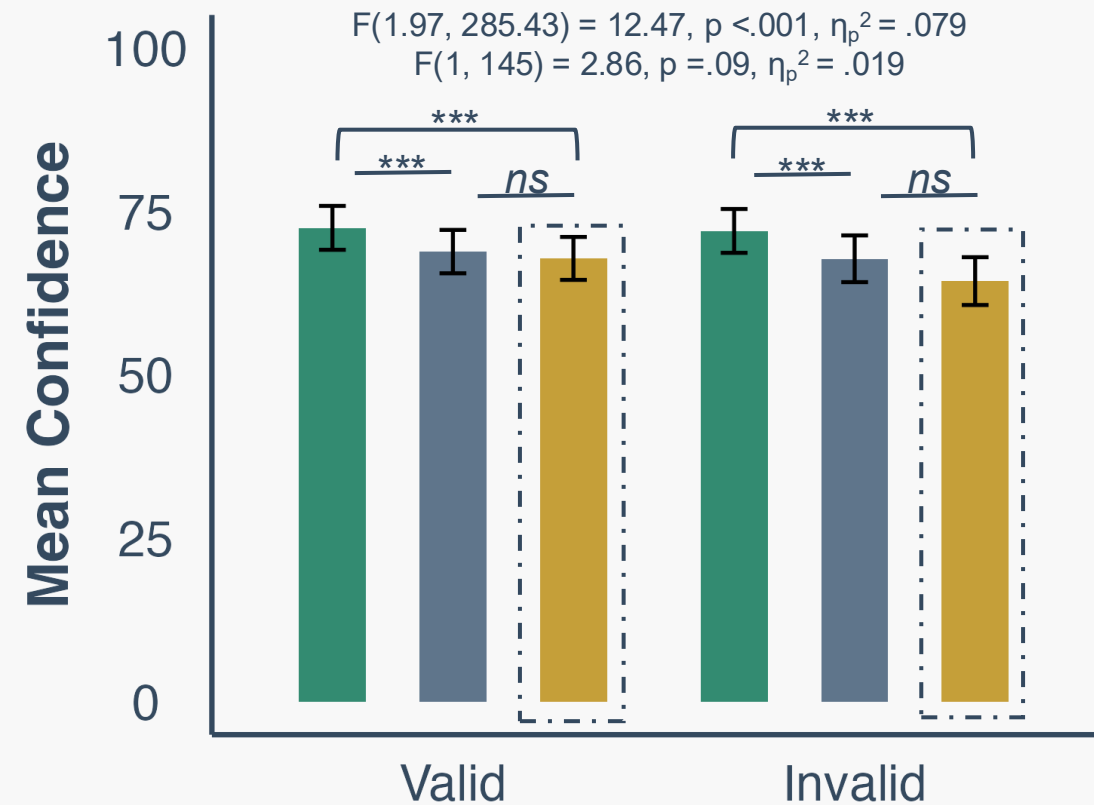
H1: Believable (easiest) < Unbelievable < Neutral (hardest)

Believable = Unbelievable < Neutral (hardest)



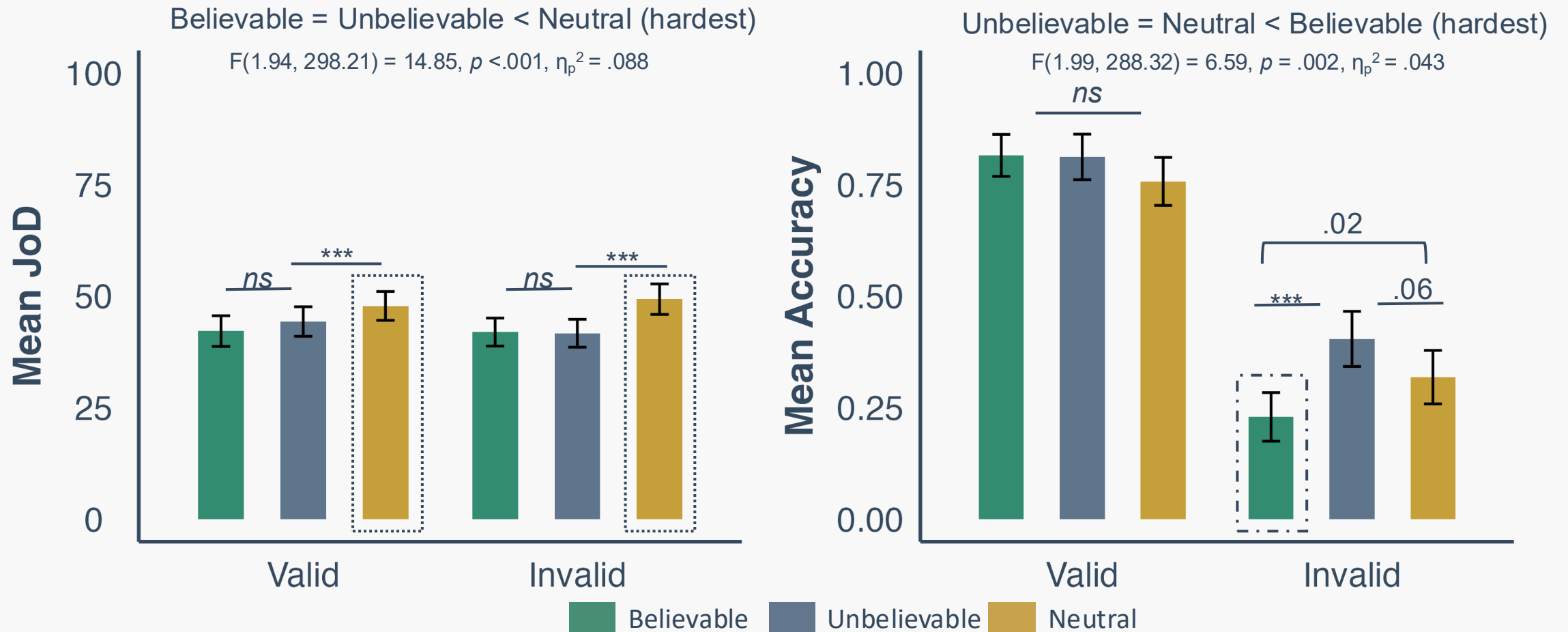
H2: Confidence tracks JODs (low difficulty → high confidence)

Believable > Unbelievable = Neutral (low confidence)



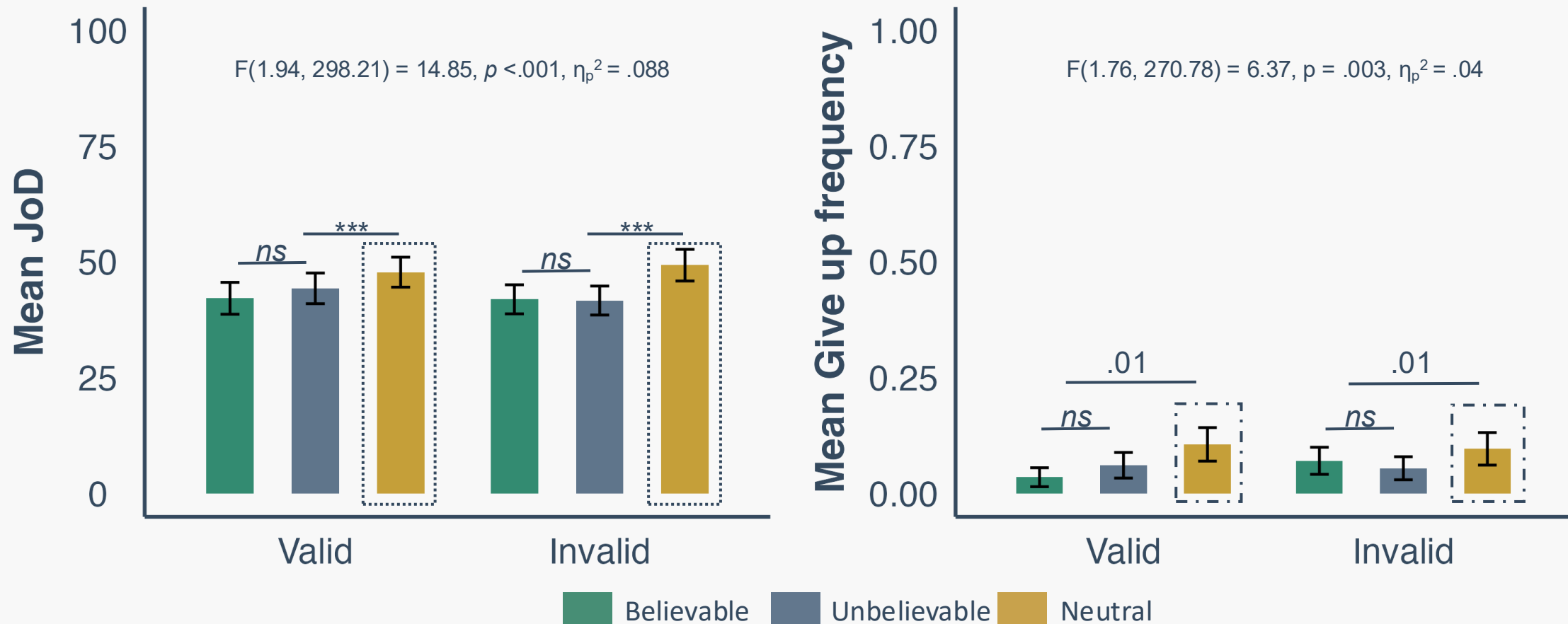
Study 2: Believability leads to metacognitive illusions

H3: JOD-Accuracy dissociation; conclusion believability did not affect JOD, but affects performance



Study 2: Believability affects control decisions

H4: The decision to give up will track JODs; people will give up more on problems with neutral conclusions



Conclusions

JODs are subject to (*misleading*) theory-based inferences in logical reasoning

- Familiarity : Believe familiar terms affect difficulty → No effect on performance
- Believability: No theories about believability → Strong effect on performance

Prospective and retrospective judgements might be sensitive to different cues

Misleading cues and inferences directly affect control decisions such as giving-up



Thank you!